ZCAS University

CAS6181 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

MID TERM TEST

21ST APRIL 2026

17: 30 – 19:00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Answer any **TWO** questions from the three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

QUESTION 1 (25 marks)

(a) Define Artificial Intelligence in your own words. According to the presentation, what four basic functions does AI typically encompass? (5 marks)

(b) Understanding the **Machine Learning analysis pipeline** is essential for building effective AI systems. Answer the following: (8 marks)

(i) List and briefly describe the **five stages** of a typical machine learning pipeline in correct sequential order. (5 marks)

(ii) What is the difference between **standardisation** and **normalisation** in data preprocessing? When would you use one over the other? (3 marks)

(c) Distinguish between the following three types of machine learning, providing one original example for each: (6 marks)

- Supervised learning
- Unsupervised learning
- Semi-supervised learning

(d) Exploratory Data Analysis (EDA) is a critical step before model training. (6 marks)

(i) What are the main goals of EDA? List at least three. (3 marks)

(ii) Name two common visualisation techniques used in EDA and explain what each reveals about the data. (3 marks)

QUESTION 2 (25 marks)

(a) An intelligent agent perceives its environment through sensors and acts through actuators. (6 marks)

(i) List and briefly describe the **four components** of an intelligent agent's architecture as presented in the slides. (4 marks)

(ii) What distinguishes an **intelligent agent** from a simple agent? (2 marks)

(b) A **Logical Agent** uses a Knowledge Base (KB) and an Inference Engine. Answer the following: (8 marks)

(i) What two types of knowledge does the Knowledge Base store? Provide an example of each. (4 marks)

(ii) What are the two key properties of a good Inference Engine? Define each. (4 marks)

(c) Compare and contrast **Forward Chaining** and **Backward Chaining** using the table format below. (6 marks)

Feature	Forward Chaining	Backward Chaining
Strategy (Data-driven or Goal-driven?)		
Starting point		
Risk / Disadvantage		
Best used for (example application)		

(d) A hospital wants to build an expert system to diagnose diseases based on patient symptoms. Would you recommend **Forward Chaining** or **Backward Chaining**? Justify your answer. (5 marks)

QUESTION 3 (25 marks)

(a) Explain the **Markov Property** using a formal mathematical expression. Provide a real-world example of a Markov process (not from the slides). (5 marks)

(b) What is a **Hidden Markov Model (HMM)**? In your answer, clearly distinguish between: (6 marks)

(i) Hidden states vs. Observable states (3 marks)

(ii) Transition probabilities vs. Emission probabilities (3 marks)

(c) An HMM is mathematically defined as $\lambda = (A, B, \pi)$. Explain what each of these three components represents. (6 marks)

(d) An HMM can solve three types of problems. Complete the following table: (8 marks)

Problem Name	What question does it answer?	Algorithm used
Evaluation Problem		
Decoding Problem		
Learning Problem		

(e) Speech recognition systems often use HMMs. Describe how hidden states and observations would be defined in this context. Additionally, state **one advantage** and **one limitation** of using HMMs for this task. (5 marks)